

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fourth Semester of B. Tech (IT) Examination****May 2015****IT212 Computer Organization & Microprocessor****Date: 04.05.2015, Monday****Time: 10:00 a.m. To 1:00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the following:****[10]**

- a. Microprocessor was introduced in which generation of Computer?
A. Second Generation B. Third Generation C. Fourth Generation D. Both B and C
- b. Fifth generation computers are based on
A. Artificial Intelligence B. Programming Intelligence C. System Knowledge D. VVLSI E. None of these.
- c. State True or False: Instruction size in CISC architecture is fixed.
- d. Global Initialized data goes to
A. Data segment B. bss segment C. code segment D. stack segment
- e. Which of the following instruction saves PC=PC+4 in to register 31.
A. j B. jr C. jalr D. jr and jalr
- f. The addressing mode/s, which uses the PC instead of a general purpose register is _____ .
A. Indexed with offset B. Relative C. direct D. both a and c
- g. The addressing mode, where you directly specify the operand value is _____ .
A. Immediate B. Register Direct C. Definite D. Relative
- h. Which of the following would correctly translate the pseudo instruction move \$t1, \$t0 into real MIPS Command?
A. add \$t1, \$t0, \$0 B. add \$t1, \$t0, \$t0 C. add \$t0, \$t1, \$t1 D. addi \$t0, \$t1, 0
- i. If an Pentium III runs a program in 8 seconds and PowerPC runs the same program in 10 seconds. How many times faster is the Pentium III?
- j. Consider the following four terms: CPU Time, Clock cycle time, Instruction count and CPI (Clock per instruction). Form an equation that relates the above four terms.

- Q – 2.a** What is Software abstraction? [02]
- Q – 2.b** Write MIPS code to multiply two numbers using “ADD” Instruction. [05]

OR

- Q – 2.b** Write MIPS code to find value of 2^i using shift instruction. [05]
- Q – 2.c** What is architecture space? List and explain the major features that differentiate various architectures? [04]

OR

- Q – 2.c** What is Assembly Language? Why do we need it? What is the function of Assembler? [04]
- Q - 3 Answer the following:(Any Two)** [14]
- Draw and explain the single cycle data path for memory reference instructions.
 - How exceptions are handled in processor data path? Elaborate with example.
 - What is pipeline hazard? Give solution of each hazard.

SECTION – II

- Q - 4 Do as directed:** [10]
- Give the difference between Hardwired and Micro-programmed Control Unit. [02]
 - What is a register window of SPARC? What is use of global/shared registers in register window? [02]
 - 1)If clock rate is 50 MHZ, find execution time for a program with 1000 instructions , if CPI for program is 3.5? [02]
 - 2)What if clock rate increases from 50MHZ to 250 MHZ and other functions remain same, how many times faster will computer be?
 - Show the IEEE 754 binary representation of the number -0.75_{10} in single precision format. [01]
 - How are instructions "sra" & "srl" different? [01]
 - Which are the possible number representation other than 2's complement? Why 2's complement is most popular? [02]
- Q – 5.a** Find the hazards in the following code segment and reorder the instructions to avoid any pipeline stalls. [03]
- ```
lw $t1, 0($t0)
lw $t2, 4($t0)
```

add \$t3, \$t1,\$t2

sw \$t3, 12(\$t0)

lw \$t4, 8(\$01)

add \$t5, \$t1,\$t4

sw \$t5, 16(\$t0)

**Q – 5.b** Carry look ahead adder is faster than Ripple carry adder. Justify. **[07]**

**OR**

**Q – 5.b** What is an activation record? When do we need it? Give an example. **[07]**

**Q – 5.c** Multiply the (-2) with (3) using Booth's algorithm. Give each step. **[05]**

**OR**

**Q – 5.c** How to interface I/O devices with processor? **[05]**

**Q – 6 Attempt any Two:** **[10]**

- a. What is a cache miss? What is Temporal and spatial locality? How does temporal and spatial locality help in reducing cache miss?
- b. How is rounding handled in floating point representation?
- c. Explain Direct Memory Access mechanism..

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